



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Fibonacci series ($N \leq 50$)

FIBONACCI SERIES ($N \leq 50$)

Develop a C program to print the first N terms of the Fibonacci series ($N \leq 50$) without using recursion. Use two variables to generate the series iteratively and store the series in an array. The program should also calculate and display the sum of the series and identify the largest term. Input validation is mandatory for N ($1 \leq N \leq 50$).

Sample Input/Output: text

Enter number of terms (1-50): 10

Fibonacci Series: 0 1 1 2 3 5 8 13 21 34

Sum of series: 88

Largest term: 34

Enter number of terms (1-50): 60

Invalid input! Please enter terms between 1 and 50.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int N;
5     int fib[50];
6     long long sum = 0;
7     int largest = 0;
8
9     while (1) {
10         printf("Enter number of terms (1-50): ");
11         if (scanf("%d", &N) != 1) {
12             printf("Invalid input! Please enter a valid integer.
13             return 0;
14         }
15         if (N < 1 || N > 50) {
16             printf("Invalid input! Please enter terms between 1
17             continue;
18         }
19         break;
20     }
21
```

OUTPUT

```
Enter number of terms (1-50): Invalid input! Please enter
terms between 1 and 50.
Enter number of terms (1-50): Fibonacci Series: 0
Sum of series: 0
Largest term: 0
```

INPUT

0 1

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